

Experiment 'p_4_3_c_replace_vowels' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_4_3_c_replace_vowels

Natural language statement: null

Method signature: p_4_3_c_replace_vowels(s: string) returns (result: string)

Ensures

- `|result| == |s|`
- forall `i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o','u','A','E','I','O','U'] then result[i] == '_' else result[i] == s[i])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_3_c_replace_vowels(s: string) returns (result: string)`

The method should respect the following contract:

ensures `|result| == |s|`, ensures forall `i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o',
↪ 'u','A','E','I','O','U'] then result[i] == '_' else result[i] == s[i])`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnLr2ePkjCYio3vBviVLX8JUvb1La

Model: gpt-5.5-2026-04-23

Created at: 1780654452

Finish reason: stop

Usage: 360 tokens in, and 1599 tokens out

```
{  
  result := seq(|s|, i => if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']  
    ↪ then '_' else s[i]);  
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_4_3_c_replace_vowels/program.dfy(5,30): Error: index out of range  
|  
5 | result := seq(|s|, i => if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']  
  ↪ then '_' else s[i]);  
  | ^
```

Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
→ Dafny body, including the curly braces that surround it. Do not show the signature
→ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnLrptA3ZIs53lDNJFJLcHpq6bzbW

Model: gpt-5.5-2026-04-23

Created at: 1780654501

Finish reason: stop

Usage: 612 tokens in, and 710 tokens out

```
{
  result := "";
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant |result| == i
    invariant forall j :: 0 <= j < i ==> (if s[j] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then result[j] == '_' else result[j] == s[j])
  {
    var c := if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then '_' else s[i];
    result := result + [c];
    i := i + 1;
  }
}
```

Final Program

```
method p_4_3_c_replace_vowels(s: string) returns (result: string)
  ensures |result| == |s|
  ensures forall i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o','u','A','E','I','O','U'] then result[i] == '_' else result[i] == s[i])
{
  result := "";
  var i := 0;
  while i < |s|
    invariant 0 <= i <= |s|
    invariant |result| == i
    invariant forall j :: 0 <= j < i ==> (if s[j] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then result[j] == '_' else result[j] == s[j])
  {
    var c := if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then '_' else s[i];
    result := result + [c];
    i := i + 1;
  }
}
```

Total Token Usage

Input tokens: 972

Output tokens: 2309

Reasoning tokens: 2026

Sum of ‘total tokens’: 3281

Experiment Timings

Overall Experiment started at 1780654451849, ended at 1780654533683, lasting 81834ms (81.83 seconds)

Iteration #1 started at 1780654451849, ended at 1780654500858, lasting 49009ms (49.01 seconds)

Iteration #2 started at 1780654500859, ended at 1780654533683, lasting 32824ms (32.82 seconds)